Multivariate Probabilistic Day-Ahead Electricity Price Forecasting and Cross-Auction Trading in the UK Market



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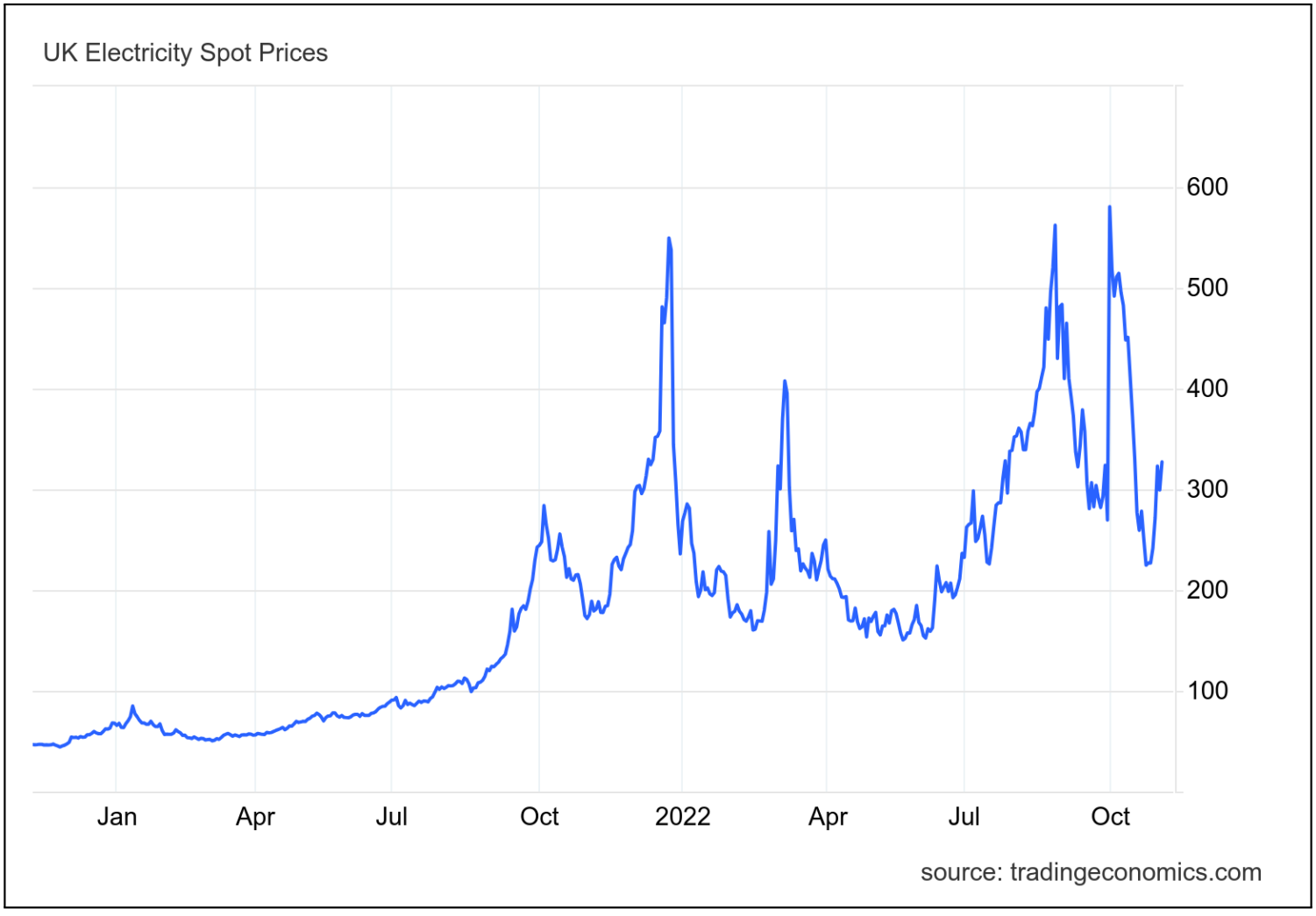


Figure 1: UK Electricity spot prices 21.01-22.10



Post-Brexit GB Power market

- Not a part of the European Single Day-ahead Coupling (SDAC)
- Two separate day-ahead 60-min auctions for GB:
  - Exchange EPEX
  - Exchange Nord Pool
- Opportunity to profit: long in one and short in the other

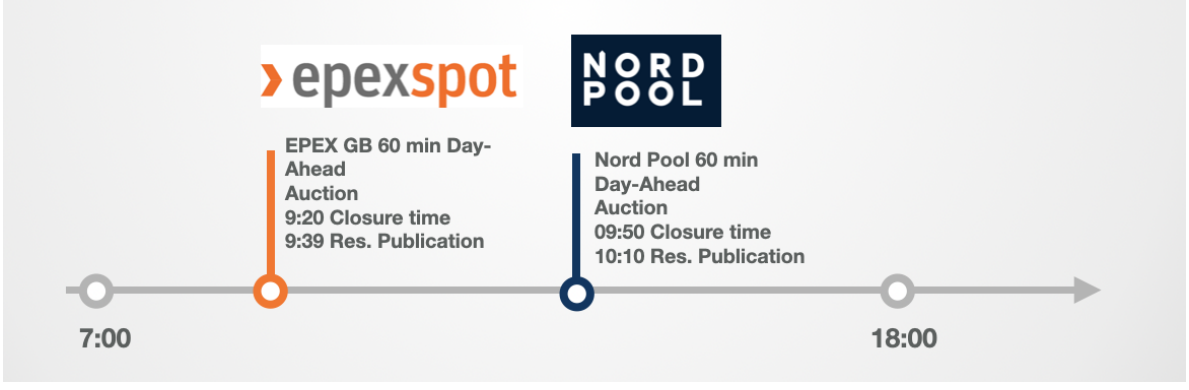
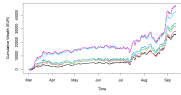


Figure 2: Nord Pool nordpoolgroup.com and EPEX epexspot.com Auctions timing



EPF Challenges

- **High Volatility and Complexity:**
  - Need for real-time balance between supply and demand.
  - High dependence on external factors like weather conditions, renewable energy sources and regulatory changes.
- **Impact of Economic and Political Events:**
  - Can lead to sudden price peaks/drops and unpredictability in electricity prices.
- **Limitations of Traditional Forecasting:**
  - Point forecasts fall short in capturing the full scope of price dynamics, making probabilistic forecasting a necessity.



Distributional Modeling

- Whole distribution needed to optimally place bids
  - Point estimate of mean: no probability of the bid being accepted
  - Risk can be incorporated, e.g. CVaR
  - Proper risk management

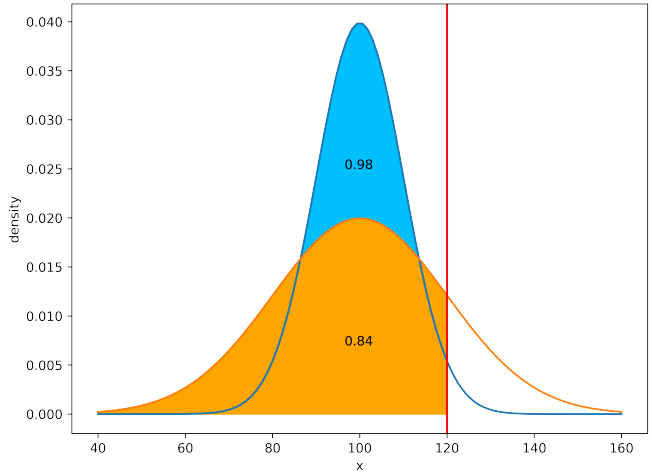
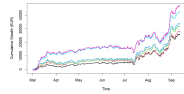


Figure 3: Two PDFs: $\mathcal{N}(100, 10^2)$ and $\mathcal{N}(100, 20^2)$. Cutoff area shows the probability of accepting a bid of 120.

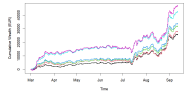


Johnson's SU in German EPF

- Application of Johnson's SU distribution in both probabilistic forecasting and EPF

	MAE	RMSE	CRPS	Kupiec 50%	Kupiec 90%
LEAR-Ens	4.372	6.375	–	–	–
DNN-Ens	3.610	5.850	–	–	–
naive	9.336	14.358	3.585	**21**	**23**
LEAR-QRA	4.161	6.676	1.575	10	8
LEAR-QRM	4.285	6.788	1.662	6	3
DNN-QRA	3.668	5.845	1.399	6	10
DNN-QRM	3.670	**5.821**	1.412	9	8
DDNN-N-pEns	3.663	5.962	1.351	2	6
DDNN-N-qEns	3.670	5.962	1.348	13	20
DDNN-JSU-pEns	**3.542**	6.146	1.304	1	4
DDNN-JSU-qEns	3.564	6.174	**1.299**	14	13

Figure 4: Forecasting accuracy on the German EPEX market; `probNN-JSU` variants lead on MAE and CRPS. Source: Marcjasz et al. (2023).



Johnson's SU: better fit for electricity prices

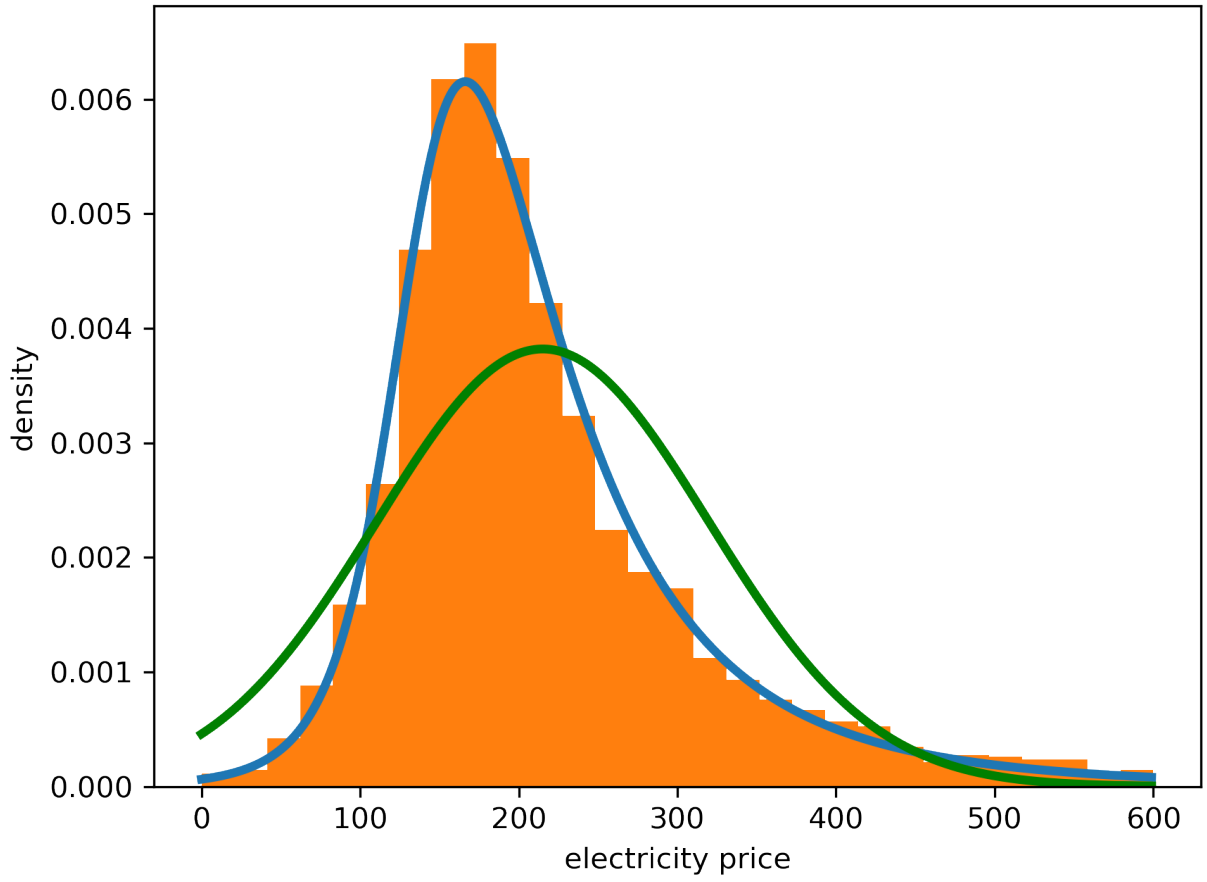
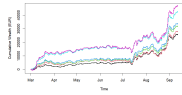


Figure 5: Johnson's SU and Gaussian fit for sample EPEX electricity prices (GBP), compared to the empirical distribution.



QQ-plot shows deviations in the tails

- Fitting on smaller time periods may be necessary
- Still, a considerably better fit than Gaussian

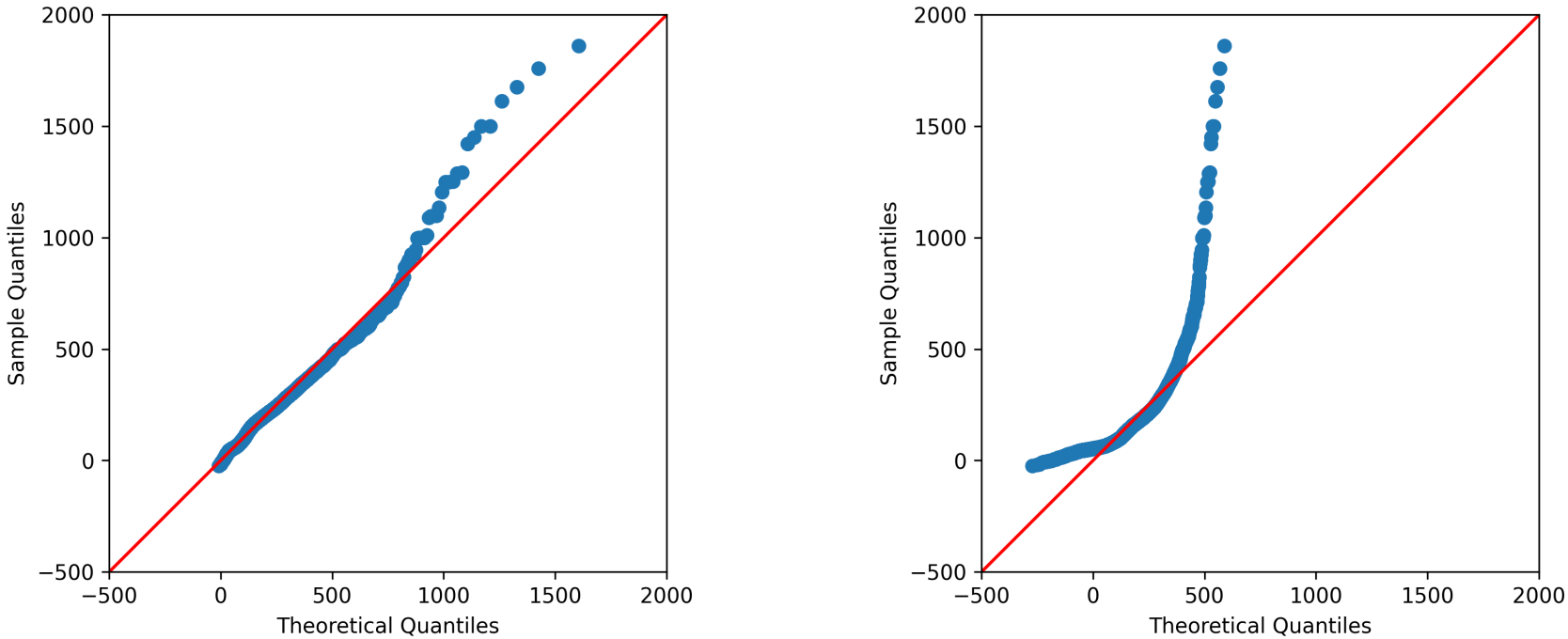
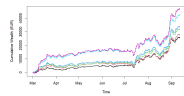


Figure 6: QQ-plot vs Johnson's SU (left) and Gaussian (right) for EPEX, 01/2021–09/2022.

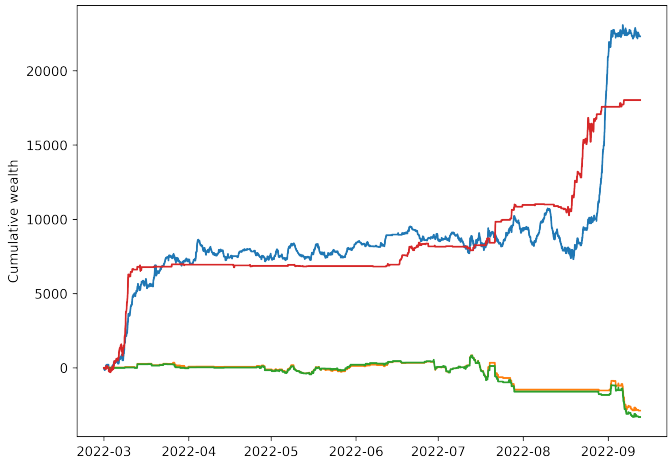


Objectives

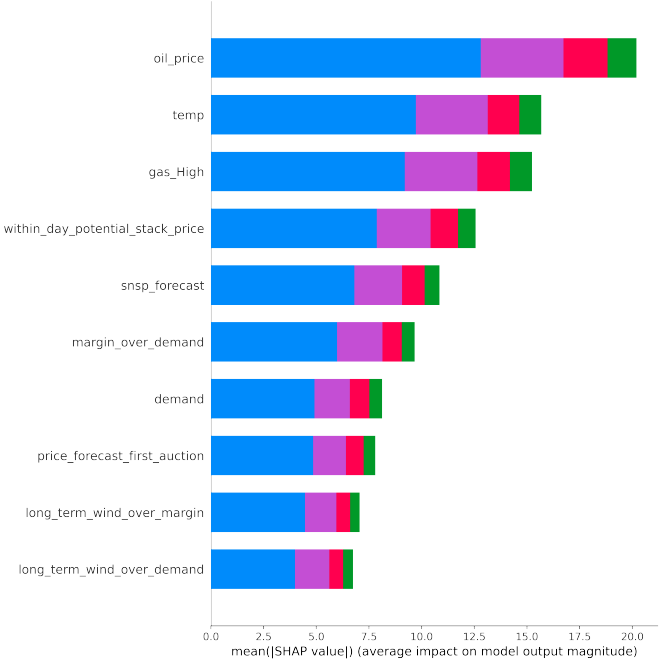
- **Probabilistic and Multivariate target modeling:**
  - Extending distributional copula regression (Hans et al., 2023).
  - Tackling complexity with Johnson's SU distribution.
- **Benchmarking against state of the art EPF approaches**
- **Economic analysis:**
  - Implementing a trading simulation on the post-Brexit UK electricity market.
  - Assessment of risk and return with Mean-CVaR strategy.
  - Risk management with a quantile bidding behavior.



DMLP joint forecasts with trading applications in UK (Agakishev et al. 2025)



Cumulative wealth, DMLP: Mean-CVaR, Mean, Profit, Median. Mar–Sep 2022.

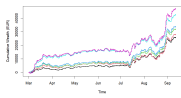


Mean SHAP values, EPEX (DMLP) — fundamentals (oil, gas, temperature) lead.



EPF State of the Art

- **From point to probabilistic EPF.** Maciejowska, Uniejewski, and Weron, 2023; Lago et al., 2021; Nitka and Weron, 2023
- **Flexible distributional learning.** Marcjasz et al., 2023; Salinas et al., 2020; Barnes and Barnes, 2021
- **Decision value for market participants.** Go beyond statistical performance and include trading, storage and risk-aware bidding evaluation, while treating economic backtests with care. Nitka and Weron, 2023; Agakishiev et al., 2025; Hirsch and Ziel, 2026
- **Multivariate dependence and scenarios.** Cross-auction, hourly and balancing-price dependence is central for coherent scenarios, stochastic programs and risk management. Agakishiev et al., 2025; Grothe, Kächele, and Krüger, 2023; Hirsch and Ziel, 2026



Outline

1. Motivation ✓
2. Methodology
3. Data
4. Empirical results
5. Conclusions



Pipeline

Phase 1: Distribution Assessment

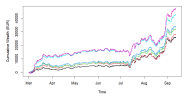
- Fit three price targets to distributions
- Evaluate fit statistically
- Identify best distributions

Phase 2: Model Selection and Forecasting

- Define forecasting model
- Refine using out-of-sample data
- Forecast future parameters

Phase 3: Trading Strategy Implementation

- Generate future scenarios
- Develop trading strategy
- Assessment/Evaluation of risk



Distribution Assessment

- **Normal Distribution:** $\mathcal{N}(\mu, \sigma^2)$
  where μ and σ are the mean and standard deviation.
- **Johnson's SU Distribution:** $\mathcal{J}(\mu, \sigma, \nu, \tau)$
  where μ and σ are location and scale parameters, while ν and τ are the shape parameters.

$$F(x) = \Phi\left(\nu + \tau \sinh^{-1}\left(\frac{x-\mu}{\sigma}\right)\right)$$

where Φ is the CDF of the standard normal distribution.

The vector of parameters will be referred as $\boldsymbol{\theta}$.



Dependence Modeling via Copulae

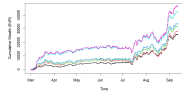
Clayton Copula

$$C(y_1, y_2, y_3; \theta_c) = \left(y_1^{-\theta_c} + y_2^{-\theta_c} + y_3^{-\theta_c} - 2 \right)^{-\frac{1}{\theta_c}}$$

Gumbel Copula

$$C(y_1, y_2, y_3; \theta_c) = \exp\left[- \left\{ (-\ln y_1)^{\theta_c} + (-\ln y_2)^{\theta_c} + (-\ln y_3)^{\theta_c} \right\}^{\frac{1}{\theta_c}} \right]$$

where $\mathbf{Y} = (y_1, y_2, y_3)$ represent the target marginal CDFs and θ_c is the copula parameter.



Structured Additive Predictors

- Each distributional parameter (Hans et al., 2023):

$$\theta_k = h_k(\eta_k) \quad \text{and} \quad \eta_k = h_k^{-1}(\theta_k),$$

where h_k^{-1} is the link function and η_k the structured additive predictor, $k \in K_1 + K_2 + K_3 + K_c$ parameter space.

- The structured additive predictor:

$$\eta_k = \beta_{0,k} + \sum_{j=1}^{J_k} f_{j,k}(x_j),$$

where $\beta_{0,k}$ is the intercept, $f_{j,k}$ are generic functions of covariates x_j and J_k the individual subset of covariates associated to θ_k.



Modeling of non-linearity

Linear effects: $f_j^{\text{linear}}(x_j) = \beta_j x_j$

Nonlinear effects: $f_j^{\text{smooth}}(x_j) = \sum_{l_j=1}^{L_j} \beta_{j,l_j} B_{j,l_j}(x_j)$

via **P-splines** with L_j B-Spline basis functions $B_{j,l_j}(x_j)$.

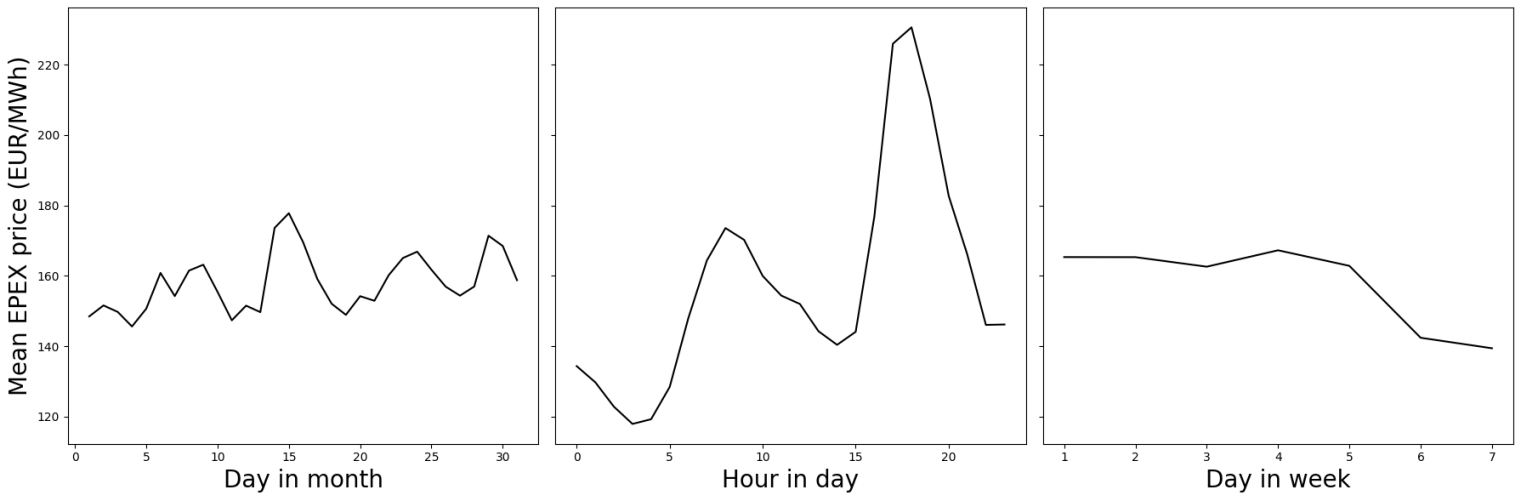
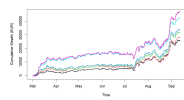


Figure 7: Monthly, daily and weekly seasonality.



Estimation via Model-Based Boosting

Parameter estimation for a trivariate target $\mathbf{Y}$ with predictors $\boldsymbol{\eta}$:

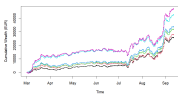
$$\hat{\boldsymbol{\eta}} = \arg\min_{\boldsymbol{\eta}} \left[\mathrm{E} \left\{ \omega \left(\mathrm{Y}; \eta_1^{(1)}, \ldots, \eta_{K_1}^{(1)}, \ldots, \eta_1^{(c)}, \ldots, \eta_{K_c}^{(c)} \right) \right\} \right],$$

where $\omega(\cdot)$ is the loss function (negative log-likelihood) and K_1, K_2, K_3, K_c the cardinality of targets and copula parameters.

Given a data sample with observations $i = 1, \ldots, n$, the empirical risk to be minimized:

$$\hat{\boldsymbol{\eta}} = \arg\min_{\boldsymbol{\eta}} \frac{1}{n} \sum_{i=1}^{n} \omega \left(\mathrm{y}_i; \eta_{(i),1}^{(1)}, \ldots, \eta_{(i),K_1}^{(1)}, \ldots, \eta_{(i),1}^{(c)}, \ldots, \eta_{(i),K_c}^{(c)} \right)$$

Boosting algorithm



Benchmark Models

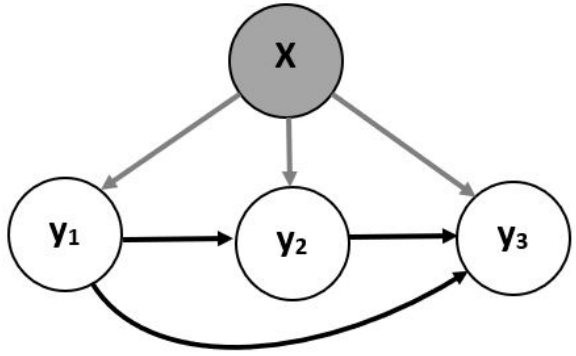
LASSO Regression:

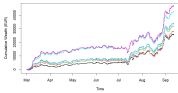
- Linear regression with a penalty term to prevent overfitting.
- Optimization problem:

$$\min_{\beta} \left\{ \frac{1}{2n} \|y - x\beta\|_2^2 + \lambda \|\beta\|_1 \right\}$$

- $\|y - x\beta\|_2^2$ is the residual sum of squares.
- $\|\cdot\|_1$ is the $\mathcal{L}1$ norm with $\lambda = 1$.

Regressor Chain: handles intra-target correlations by sequentially predicting targets.





Benchmark Models

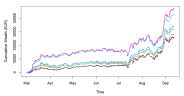
Regressor Chain Frameworks:

- Permutation Scheme (Geiß et al., 2022): considers all permutations of the order, resulting in 6 point forecasts.
- Forecast Averaging (Lago et al., 2021): forecasts from different rolling calibration windows (84, 212, 424 days) resulting in 3 point forecasts.

Quantile Regression Averaging (Nowotarski and Weron, 2015):

- Point forecasts as independent variables and the corresponding observed price as the dependent variable.
- Weighted combination of the forecasts via minimization of Pinball Score quantilewise.

Statistical Performance Metrics



Trading Strategy 1

Two UK electricity auctions: EPEX (y_1) and Nord Pool (y_2).

Objective: buy (or sell) at a lower (higher) price in the first auction and sell (buy) at a higher (lower) price in the second auction. All left after the second auction is settled by system prices (y_3).

Trading setting:

- Volume: 1 MWh per hour
- Simulation: 1000 scenarios from forecasted distribution
- Aggregation schemes:
  - Mean: average performance across scenarios.
  - Mean-CVaR: Mean - $\gamma \cdot$ CVaR , with CVaR at 0.95 level and $\gamma = 0.3$.

Bids Optimization



Trading Strategy 2

$$\Pi_d = -\frac{1}{0.9}\hat{Y}^{0.5}_{d,h_1,a} + 0.9\hat{Y}^{0.5}_{d,h_2,a} - 1_{\{B=0\}}\frac{1}{0.9}\hat{Y}^{0.5}_{d,h^*,a^*} + 1_{\{B=2\}}0.9\hat{Y}^{0.5}_{d,h^*,a^*}$$

where Π_d is profit for day d of test set, hours h_1, h_2, h^*, auction $a = 1, 2$ (EPEX and Nord Pool), a^* includes system price, $\hat{Y}^{0.5}$ median price forecast.

Objective: buy electricity at hour h_1 and charge the battery, sell electricity at hour h_2 (where $h_2 > h_1$). Use system price for trading if necessary.

Battery:

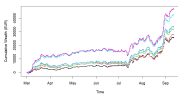
- Total usable capacity: $B = 2$ MWh (initially $B = 1$)
- Charging/Discharging efficiency: 90%



Bids Optimization

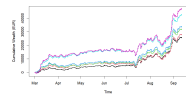
- Select hours h_1, h_2, h^* and auction a (or a^*) to maximize Π_d.
- Simple selection when the battery is halfway charged ($B = 1$), linear programming when $B \neq 1$.
- Profitability check: $-\frac{1}{0.9}\hat{Y}^{0.5}_{d,h_1,a} + 0.9\hat{Y}^{0.5}_{d,h_2,a} > 0$
- Place buy and sell orders based on quantiles $q = \frac{1+\alpha}{2}$ and $\hat{Y}^{q}_{d,h_1,a}$, $\hat{Y}^{1-q}_{d,h_2,a}$.
- Unlimited bid placed at h^* when necessary:

$$\begin{cases} h^* < h_1 & \text{if } B = 2 \\ h^* < h_2 & \text{if } B = 0 \end{cases}$$



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Data

Observations are collected on an **hourly** base.

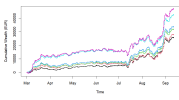
Cohort period: **01.01.2021 - 12.09.2022** (619 days, $\approx 15k$ obs.)

- Training set: 01.01.2021 - 28.02.2022 (424 days)
- Test set: 01.03.2022 - 12.09.2022 (196 days)

Distributional Copula regression model is re-trained **monthly**.

Benchmark Calibration Windows:

- LASSO + Permutation Scheme: 424 days
- LASSO + Calibration Window: 84, 212, 424 days
- Quantile Regression Averaging: 50 days from test set



Input Features:

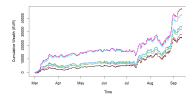
- Electricity auction prices, demand (Source: suena)
- Solar and wind generation predictions (Source: suena)
- Brent oil prices (Source: FED/FRED)
- Gas and emissions futures (Source: investing.com)
- Weather data (Source: Copernicus Climate Change Service)
- Other features:
  - Previous day prices
  - Weekday, hour of day (sine and cosine)
  - Weather anomaly

Outputs: parameters for first auction, second auction, system price distributions and copula parameter.



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Fitting results

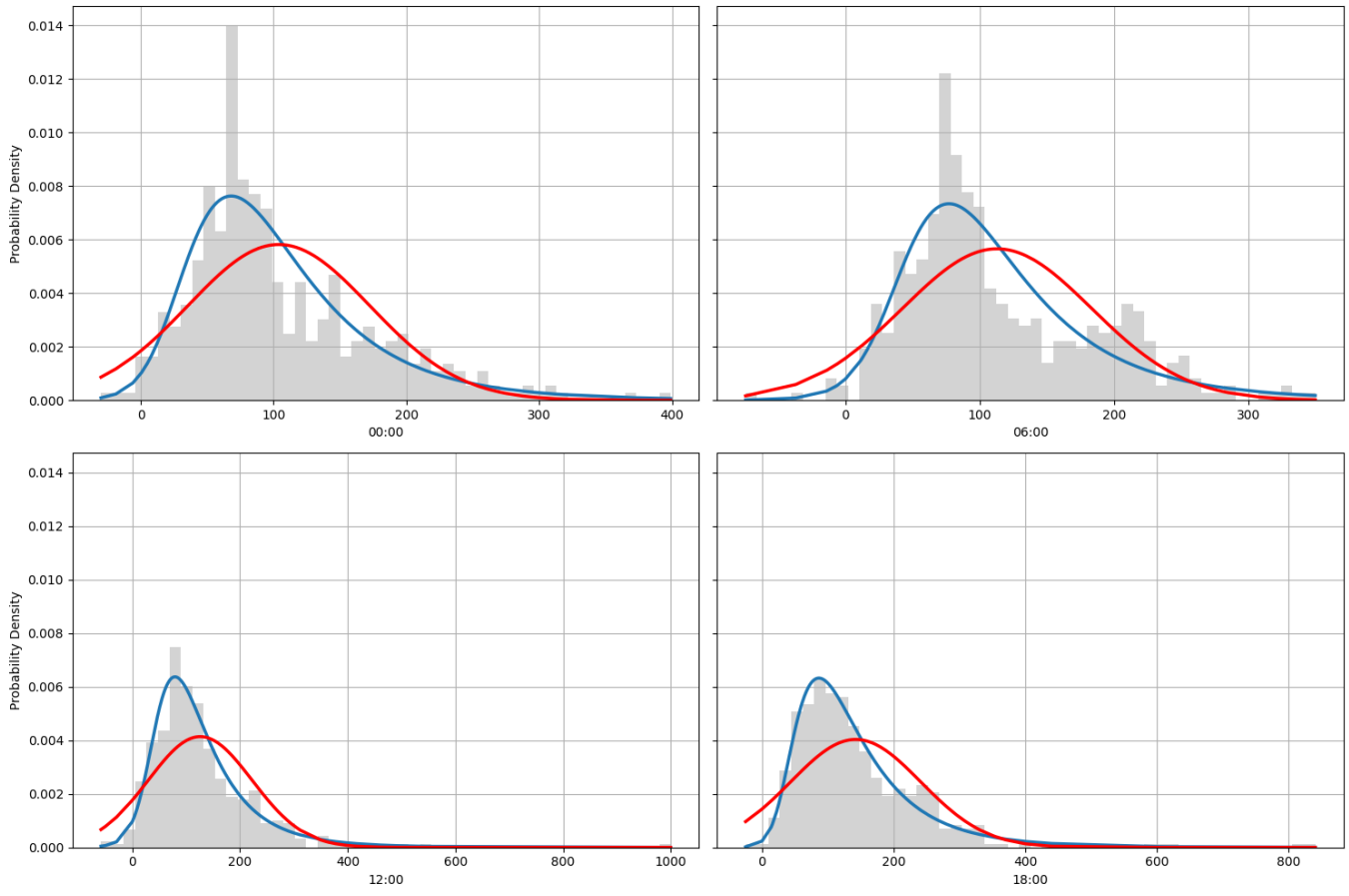
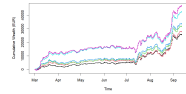


Figure 8: Realizations of System Prices at four distinct hours with fitted distributions Normal and Johnson's SU on training set



	Hour 0		Hour 6		Hour 12		Hour 18	
	Normal	JSU	Normal	JSU	Normal	JSU	Normal	JSU
CRPS	59.57	59.76	67.32	67.87	70.75	70.19	154.7	133.9
KS-test critical-value	0.064	0.064	0.064	0.064	0.064	0.064	0.064	0.064
KS-test p-value	0.001	0.050	0.000	0.020	0.001	0.044	0.000	0.005
CRPS	60.47	60.29	69.22	69.89	71.56	71.24	149.7	129.3
KS-test critical-value	0.064	0.064	0.064	0.064	0.064	0.064	0.064	0.064
KS-test p-value	0.001	0.046	0.000	0.008	0.001	0.033	0.000	0.010
CRPS	72.70	71.21	75.45	75.27	98.69	90.55	101.4	92.62
KS-test critical-value	0.064	0.064	0.064	0.064	0.064	0.064	0.064	0.064
KS-test p-value	0.000	0.272	0.000	0.218	0.001	0.774	0.001	0.725

Table 1: Combined CRPS, KS-test Critical values and p-values for EPEX auction, Nord Pool auction, and System prices across different hours and Normal vs Johnson's SU distributions



Model Selection

Copula	Marginal	Initialization	Scaling	Iteration	Risk
Clayton	JSU	No	Centered	1575	3509
Clayton	JSU	No	Non-Centered	973	694
Gumbel	JSU	No	Centered	1088	8263
Gumbel	JSU	No	Non-Centered	911	5135
Clayton	JSU	Yes	Centered	1702	2047
Clayton	JSU	Yes	Non-Centered	1094	525
Gumbel	JSU	Yes	Centered	866	5164
Gumbel	JSU	Yes	Non-Centered	209	3057
Clayton	NOR	No	Non-Centered	1411	1096

Table 2: Comparison of different Copulae, Marginal distributions, initialization and scaling by means of Predictive Risk and associated iteration of minimum risk.



Forecasting Results

	MAE	RMSE	CRPS
EPEX auction			
CLA-COP	47.884	74.825	16.852
LASSO-QRA-PS	27.926	37.839	32.605
LASSO-QRA-CW	31.318	43.484	32.213
Nord Pool auction			
CLA-COP	48.951	74.915	17.821
LASSO-QRA-PS	30.004	40.379	32.438
LASSO-QRA-CW	33.082	45.669	32.311
System price			
CLA-COP	79.858	113.85	28.656
LASSO-QRA-PS	69.679	88.889	33.036
LASSO-QRA-CW	72.528	93.192	32.973

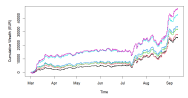
Table 3: Performance metrics for different models and target variables.



Trading Results 1

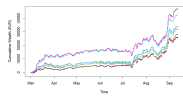
	Mean	Mean-CVaR
CLA-COP	17936	22001
LASSO-QRA-PS	9954.9	6283.3
LASSO-QRA-CW	-2623.9	-3005.1

Table 4: Cumulative wealth in EUR for different models and aggregations with main strategy.



Model	Strategy	MR	VAR	SR
CLA-COP	Mean	0.02052	0.02736	0.12405
	Mean-CVaR	0.02638	0.03464	0.14175
LASSO-QRA-PS	Mean	-0.00335	0.02816	-0.01995
	Mean-CVaR	-0.00604	0.02746	-0.03648
LASSO-QRA-CW	Mean	-0.01426	0.03331	-0.07811
	Mean-CVaR	-0.01461	0.03316	-0.08024

Table 5: Mean Return (MR), Variance (VAR), and Sharpe Ratio (SR) for different models and strategies.



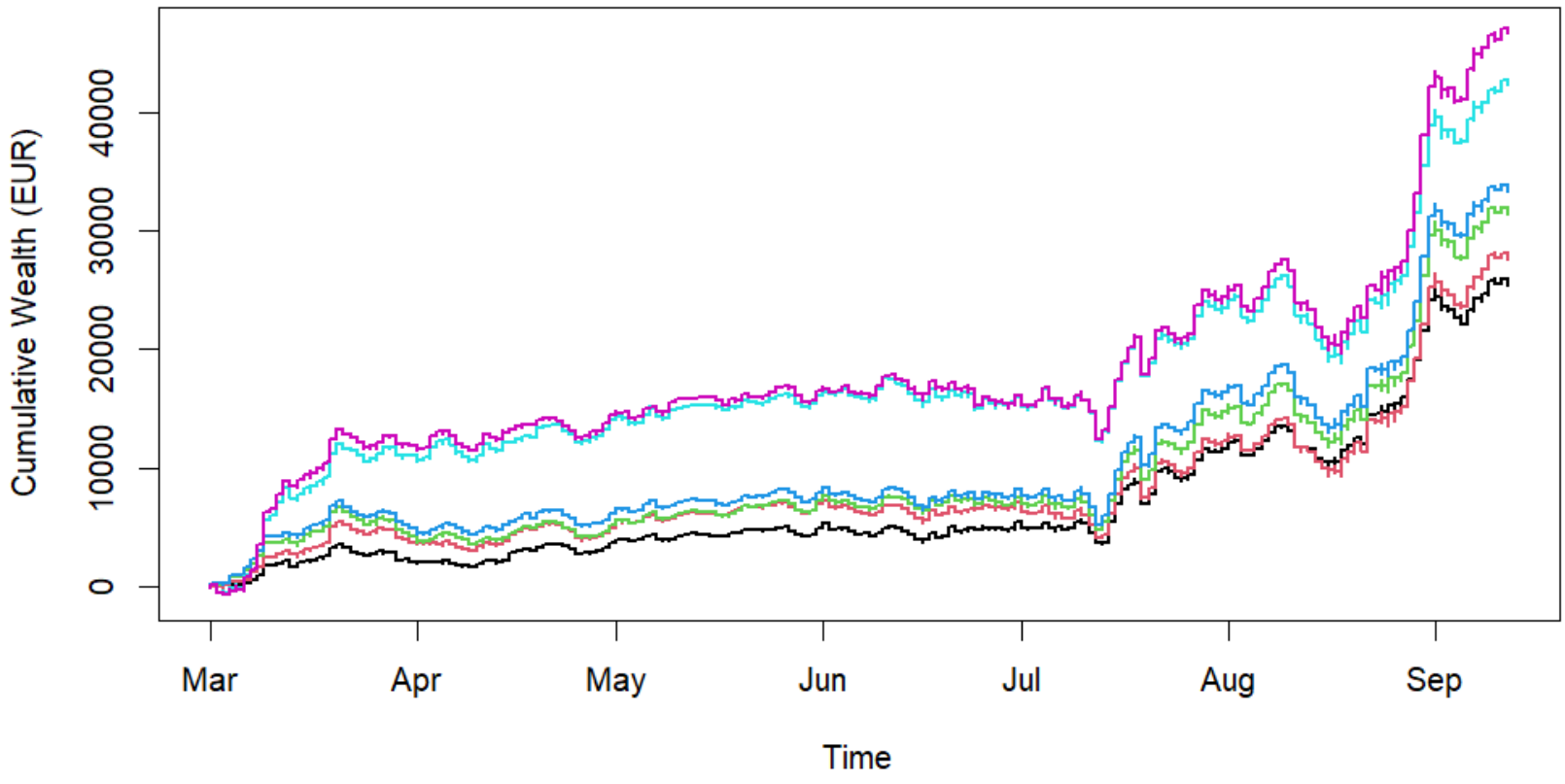
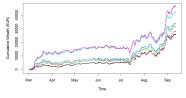


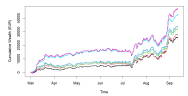
Figure 9: Cumulative wealth for $\gamma =$ **0.03**, **0.06**, **0.09**, **0.1**, **0.3**, **0.5** in Mean-CVaR strategy for Clayton Copula model.



Trading Results 2

	Alpha				
	0%	20%	40%	60%	80%
CLA-COP	346.38	831.38	1643.47	2732.85	3637.52
LASSO-QRA-PS	3939.85	3356.84	2969.82	3271.55	3688.17
LASSO-QRA-CW	1473.56	924.13	489.92	-112.75	-635.32

Table 6: Cumulative wealth in EUR for different models and alphas with benchmark strategy.



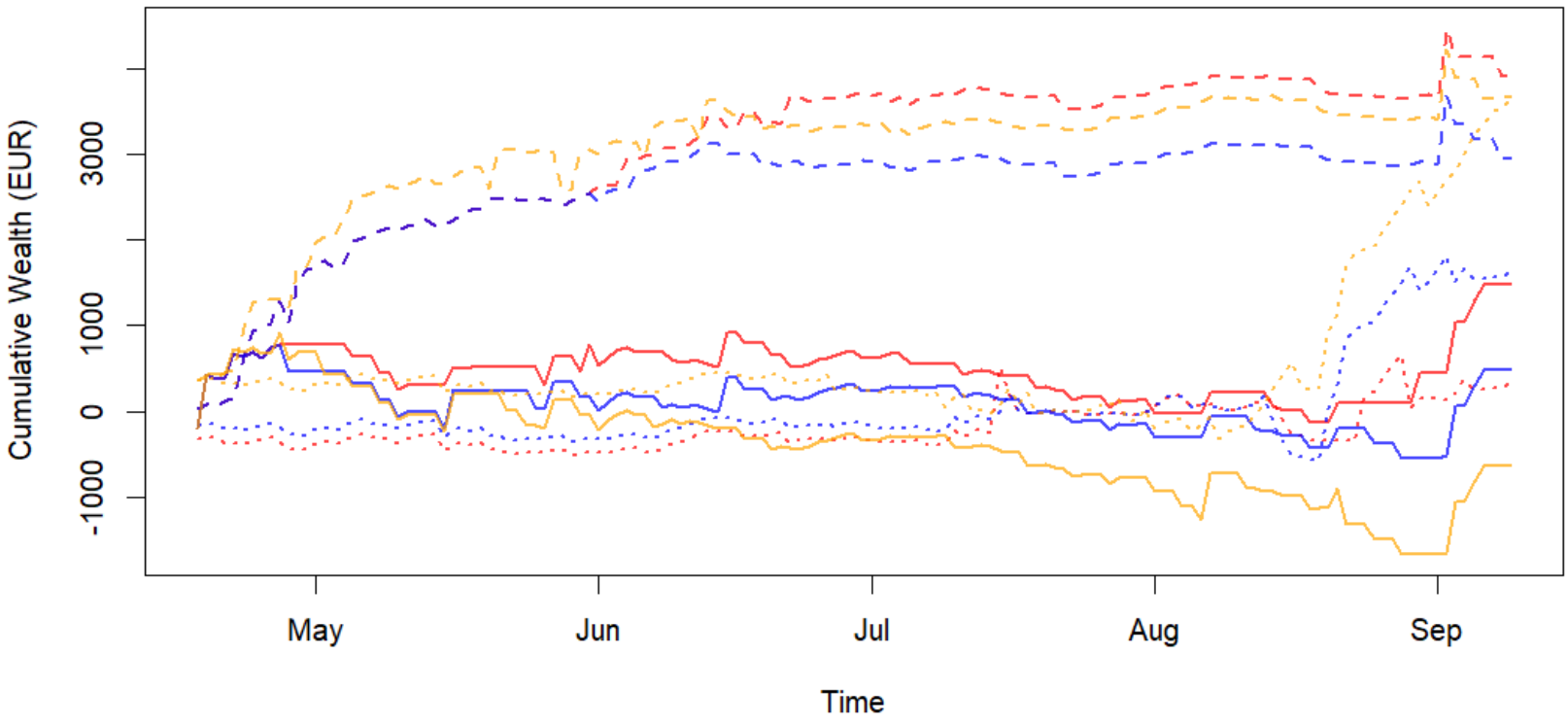
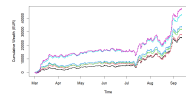


Figure 10: Cumulative wealth for $\alpha =$ **0%**, **40%**, **80%** and models in benchmark strategy. Clayton Copula (- -), Lasso Permutation Scheme (··), Lasso Calibration Window (—).



Outline

1. Motivation ✓
2. Methodology ✓
3. Data ✓
4. Empirical results ✓
5. Conclusions



Conclusions

- Extended distributional copula regression framework to a multivariate setting for electricity price forecasting.
- Clayton copula combined with Johnson's SU distribution effectively captures skewness, right-tail behavior and intra-target dependencies.
- The Clayton copula model outperforms LASSO-based models in terms of CRPS and risk-adjusted trading strategies.
- Demonstrated promising results across various risk-adjusted trading scenarios using the proposed model.



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Boosting Algorithm

1. Sequentially minimize empirical risk by gradient descent in function space.
2. The algorithm begins with an initial specification of the additive predictors and base-learners.
3. Each iteration m=1,. . .,m_{stop}, parameter-specific base learners j are fitted one-by-one to the respective negative partial gradient vectors.
4. Best base learner to fit the corresponding gradient vector is selected for each parameter θ_k (variable selection).
5. Gradients are standardized by diving the median absolute deviation (MAD) (Hofner, Mayr, and Schmid, 2016):

$$\text{MAD} = \text{median}_m \left(|\nabla_{k,m} - \text{median}_j(\nabla_{k,j})| \right),$$

where $\nabla_{k,m}$ is the gradient of the k-th parameter.



Hyperparameters

Step Length s:

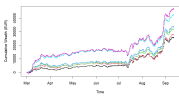
- Controls the proportion of the base-learner fit added each m.
- Ensures stability and shrinks parameter estimates toward zero.
- Higher s can foster the parameter update process.

Stopping Iteration m_{stop}:

- Determines when to cease updates, controlling model complexity.
- Early stopping selects informative variables and avoids overfitting.

Empirically, m_{stop} is then identified as the iteration with the lowest average predictive risk while a step length of ≈ 0.01, is suggested to be generally effective by simulations.

← Back to estimation



Statistical Performance Metrics

Point Forecasts:

- Mean Absolute Error (MAE): measures the average magnitude of errors in forecasts.

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

- Root Mean Squared Error (RMSE): measures the average magnitude of errors with a higher weight on larger errors.

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

← Back to benchmarks



Probabilistic Forecasts:

- Pinball Score: evaluates the accuracy of quantile forecasts.

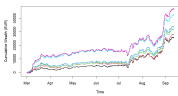
$$\text{Pinball}_\tau(\hat{Y}_{t,\tau}, Y_t) = \begin{cases} (1-\tau)(\hat{Y}_{t,\tau} - Y_t) & \text{if } Y_t < \hat{Y}_{t,\tau}, \\ \tau(Y_t - \hat{Y}_{t,\tau}) & \text{if } Y_t \geq \hat{Y}_{t,\tau}. \end{cases}$$

- Continuous Ranked Probability Score (CRPS): compares forecast distributions with observed values. We approximate by computing the average Pinball Score across 99 percentiles:

$$\text{CRPS}_{d,h} = \frac{1}{99} \sum_{q=1}^{99} \text{PS}(\hat{Y}^q_{d,h}, Y_{d,h}, q),$$

where PS is the Pinball Score for the q-th quantile.

← Back to benchmarks



Bids Optimization

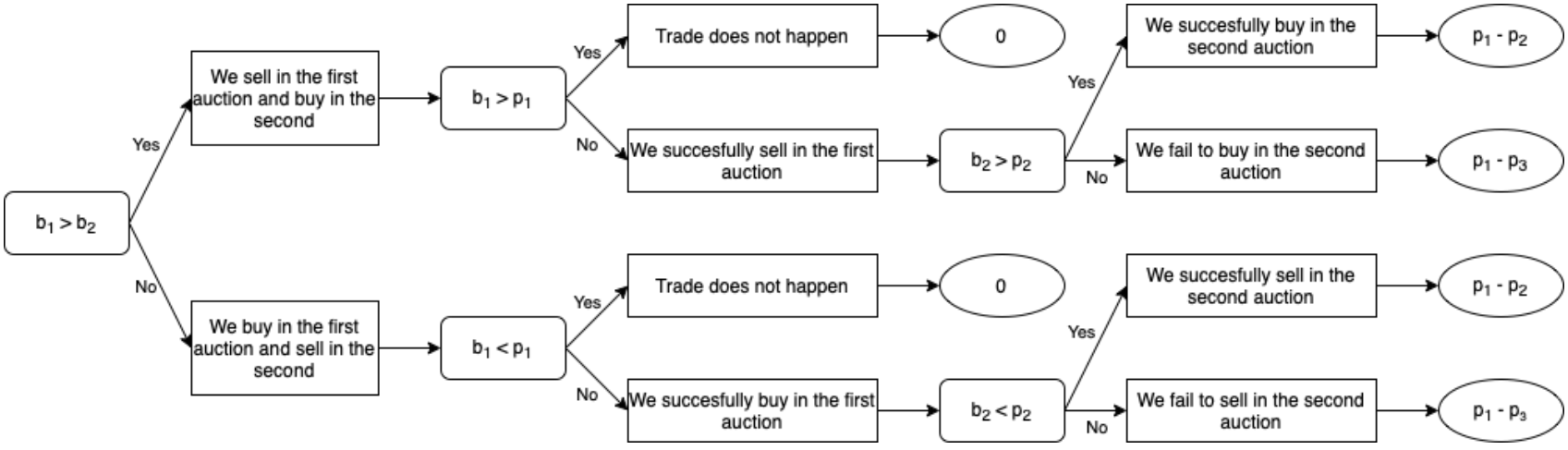


Figure 11: Bidding Schema (Agakishiev et al., 2025)

- b_1 and b_2 bids at EPEX and Nord Pool, p_1 and p_2 corresponding sampled prices, p_3 system price.
- Grid of 41 bid values from 0 to 600.
- Optimize bids to maximize performance based on chosen aggregation strategy.